Applied Physics Letters

Operation of YBCO kinetic-inductance bolometers for outer solar system missions

Cite as: Appl. Phys. Lett. **114**, 132601 (2019); https://doi.org/10.1063/1.5089143
Submitted: 16 January 2019 . Accepted: 14 March 2019 . Published Online: 01 April 2019

A. Chakrabarty, M. A. Lindeman, B. Bumble, A. W. Kleinsasser, W. A. Holmes, and D. Cunnane

View Online Export Citation CrossMark

© 2019 Author(s).

Applied Physics Letters ARTICLE scitation.org/journal/apl

Operation of YBCO kinetic-inductance bolometers for outer solar system missions

Cite as: Appl. Phys. Lett. **114**, 132601 (2019); doi: 10.1063/1.5089143
Submitted: 16 January 2019 · Accepted: 14 March 2019 ·
Published Online: 1 April 2019

View Online Export Citation CrossMark

A. Chakrabarty, M. A. Lindeman, B. Bumble, A. W. Kleinsasser,[a)] W. A. Holmes, and D. Cunnane

AFFILIATIONS

Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California 91109, USA

[a)]Author to whom correspondence should be addressed: alan.kleinsasser@jpl.nasa.gov

ABSTRACT

We developed superconducting kinetic inductance bolometers (KIBs) for hyperspectral imaging in outer solar system missions. Each bolometer is a microwave L-C resonator that will act as an individual pixel in the focal plane of an imaging Fourier transform infrared spectrometer. Our devices are fabricated from high temperature superconductor yttrium barium copper oxide (YBCO), with a superconducting transition temperature of ∼85 K. The inductors are suspended on thin Si_3N_4 membranes. Each resonator operates at a unique resonant frequency, and hence, many can be multiplexed on a single feed line using a large number of carrier frequencies. The resonators function as detectors because their resonant frequencies decrease with increasing temperature as the bolometers absorb incident radiation. In this paper, we discuss the thermal properties of a YBCO KIB and its response to signal power to verify that the response is near theoretical predictions.

Published under license by AIP Publishing. https://doi.org/10.1063/1.5089143

Kinetic inductance detectors (KIDs) have attracted significant attention for well over a decade.[1–5] Yttrium barium copper oxide (YBCO)-based kinetic inductance bolometers (KIBs)[3] are a new class of high temperature superconducting (HTS) detectors that function as bolometers,[6,7] as do their low temperature cousins, thermal KIDs (TKIDs).[8] KIDs, TKIDs, and KIBs all consist of an L-C resonator coupled to a microwave feedline. However, in a KIB, the inductor is suspended on a membrane in order to thermally isolate it from the substrate, as shown in Fig. 1. The substrate acts as a heat sink and maintains a constant bath temperature, T_b. Each KIB has a unique resonant frequency, determined by the capacitor size, and acts as a single pixel of the detector array. Many KIBs can be multiplexed on a single feedline, the number depending on the quality factor (Q) of the resonators and the bandwidth of the detectors. All the KIBs are read out simultaneously by sending a comb of frequencies through the feedline.

The main absorber of the incident radiation is YBCO. Si_3N_4 which forms the membrane contributes very little to the absorption in the wavelength range of our interest. The bolometer absorbs radiation and thermalizes the energy, creating quasiparticles (QPs) in the YBCO. This increases its kinetic inductance and therefore shifts the L-C resonant frequency. QP recombination generates phonons, producing more QPs, which increases the effective QP lifetime on the membrane. However, the QP lifetime in YBCO is short enough to ensure rapid thermalization of the membrane, assisted by the weak thermal link with the substrate. This mode of operation suppresses the internal thermal fluctuation noise due to quasiparticle generation-recombination on the membrane. Instead, thermal noise is dominated by phonon exchange between the membrane and the substrate, given by[9] $P_{ph} = \sqrt{4k_B T^2 G}$, where G is the thermal conductance of the legs that act as a weak thermal link between the membrane and the substrate.

In this paper, we characterize a single bolometer (pixel) and report its performance as a function of temperature up to 24 K. By

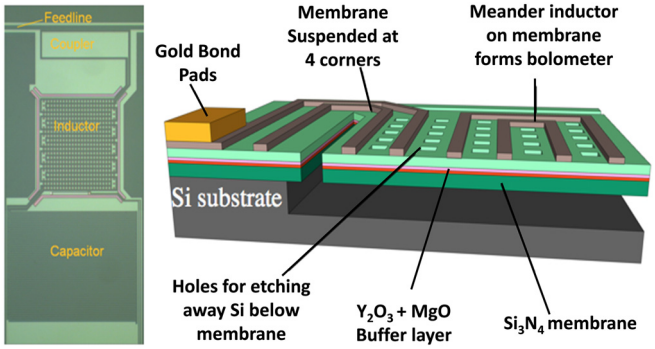

FIG. 1. Left: Optical micrograph of a KIB. The device is inductively coupled to the feedline. The inductor is suspended on a membrane by four legs in the corners. Right: 3D illustration of the part of the inductor suspended on a Si_3N_4 membrane grown on the Si substrate. The holes between the inductor segments are photolithographically patterned to etch Si beneath the membrane using XeF_2.

Published under license by AIP Publishing

measuring the full complex transmission and using the resonant circle method,[10,11] we measured the KIB parameters such as the internal quality factor Q_i, the coupling quality factor $\widehat{Q_C}$, and the resonance frequency f_r. By applying a small sinusoidal signal to the on-membrane heater, the bolometer characteristics, including heat capacity, thermal conductance, thermal time constant, and responsivity, are determined and compared with the AC bolometer theory with good agreement, at different bath temperatures. Our measurements and theoretical modelling will allow us to design YBCO KIBs with high sensitivity, optimized for high temperature operation. For planetary science missions, KIBs can be passively cooled (without a cryocooler) in outer space to an acceptable operating temperature of 55 K, which is well below the YBCO transition temperature, T_c of 85 K.[12] Because KIBs are temperature sensors, they can have very large bandwidths in the mid- to far-IR range. Therefore, these detectors, coupled with high spectral resolution dispersive optics such as Fourier transform infrared (FTIR) spectrometers, could be used for imaging the surface of outer solar system planets with enhanced spectral and spatial resolutions. Furthermore, broadband detectors can be used to study the thick atmospheres of giant planets and their moons (e.g., Saturn, Titan, Uranus, or Neptune), which requires a wide range of wavelengths to probe at different depths into their atmospheres.[13–15]

The KIBs are fabricated on a Si substrate. First, a 1 μm thick layer of Si_3N_4 is grown on Si. This will eventually form the membrane for the YBCO inductor. Since YBCO does not grow on amorphous layers like Si_3N_4,[16] a polycrystalline buffer layer consisting of Y_2O_3 (6–8 nm) and MgO (5–10 nm) is deposited on Si_3N_4 using ion beam assisted deposition (IBAD).[12] This is followed by deposition of epitaxial MgO (50 nm) at 500 °C. YBCO (50 nm) and Au (300 nm) are then successively deposited. The YBCO feedline, coupler, inductor (15 μm wide), and interdigitated capacitor (2.5 μm wide) are lithographically patterned and etched. The inductor is located on the Si_3N_4 membrane, which is connected to the substrate by four thin legs (5–15 μm wide, 60 μm long) forming a weak thermal link with the substrate, illustrated in Fig. 1. To fabricate the membrane, a grid of holes between the meandering inductor lines is defined by photolithography. The MgO, Y_2O_3, and Si_3N_4 layers are then selectively removed using a BCl_3 inductively-coupled plasma etch. In the final step, the underlying Si substrate is isotopically etched using XeF_2 to form the free-standing membrane. To calibrate our device and demonstrate its performance, a thin ~300 Ω PdAu heater is fabricated on the membrane.

The KIBs are placed in an enclosed box inside an IR-Labs Dewar, cooled to 4.2 K. The box temperature is maintained using a proportional integral derivative-controller. We use a homodyne set-up to measure the full complex transmission of the KIBs. Figure 2(a) shows the transmission coefficient S_{21} as a function of sweep frequency f, for a typical KIB resonance. The shape of S_{21} depends on the dissipation within the KIB given by Q_i and $\widehat{Q_C}$ which accounts for the coupling between the resonator L-C circuit and the feedline[17] and also for the impedance mismatch. $\widehat{Q_C}$ is a complex quantity given by $\widehat{Q_C} = Q_c \exp(i\phi)$, where ϕ is the phase lag. The loaded quality factor Q is given by $Q^{-1} = Q_i^{-1} + Q_c^{-1}$. S_{21} can be modeled by the equation[10,11]

$$S_{21} \approx a e^{\kappa} e^{-2\pi i f \zeta} \left[1 - \frac{(Q/|Q_c|)e^{i\phi}}{1 + 2iQ\left(\frac{f}{f_r} - 1\right)} \right], \tag{1}$$

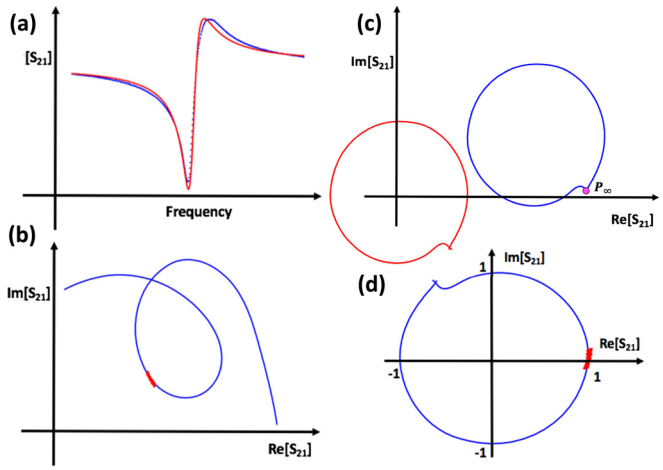

FIG. 2. Measurements of a typical KIB resonance. (a) S_{21} vs frequency (blue), fitted using Eq. (1) (red). (b) The full complex transmission, including cable delays. A small heater signal applied to the KIB biased near resonance is shown in red. (c) Complex transmission corrected for cable delay (blue). The magenta dot is the point P_∞. The resonant circle translated to the origin is shown in red. (d) The unit resonant circle, rotated so that the bias frequency point (near resonance) lies on the real axis. The calibrated S_{21}, with a small heater signal applied to the KIB and biased near resonance is shown in red, subtending an angle θ at the center of the unit resonant circle.

where ae^{κ} is the total amplification/attenuation due to amplifiers and the cable losses, $e^{-2\pi i f \zeta}$ is the cable delay owing to the finite signal speed in the cables, and ζ is the electronic delay time at a probe frequency f.

Figure 2(b) shows the real and imaginary parts of the uncalibrated complex transmission signal S_{21}. Correcting for the cable delay, the modified S_{21} looks like a circle, (resonant circle) where the point P_∞ represents the response far from resonance, Fig. 2(c) (blue). The kinks and deviations from an exact circle are due to the impedance mismatch and crosstalk in the cryogenic readout system and bias wires. The normalized resonant circle is translated to the origin and rotated so that the bias frequency (which is chosen to be near resonance) lies on the real axis, as shown in Fig. 2(d). This normalized representation is used in modeling KIBs because it removes incidental effects such as cable delay and attenuation.

The bolometer is next biased near its resonance frequency (f_r), and a small, low frequency (f) heater signal (3 V_{pp}, 2 Hz) was applied to the on-membrane heater that produced a thermal signal at double the applied frequency (4 Hz). The sinusoidal modulation of the bolometer membrane temperature mixes the low frequency (f) thermal signal with the high frequency (f_r) bias signal. Absorption of the small power ΔH alters the membrane temperature by an amount ΔT, changing the kinetic inductance of the bolometer and leading to a detectable change in the complex transmission from the steady state bias point. The perturbation from the bias point is plotted on the frequency swept full complex transmission of the KIB in Fig. 2(b) (red). The calibrated transmission due to the heater signal plotted on the unit-resonant circle in the normalized coordinates is shown in Fig. 2(d). The heater signal slightly deviates from the circular trajectory as Q alters with the change in the membrane temperature. At bath temperature T_b, the response of a bolometer to a small on-membrane

Published under license by AIP Publishing

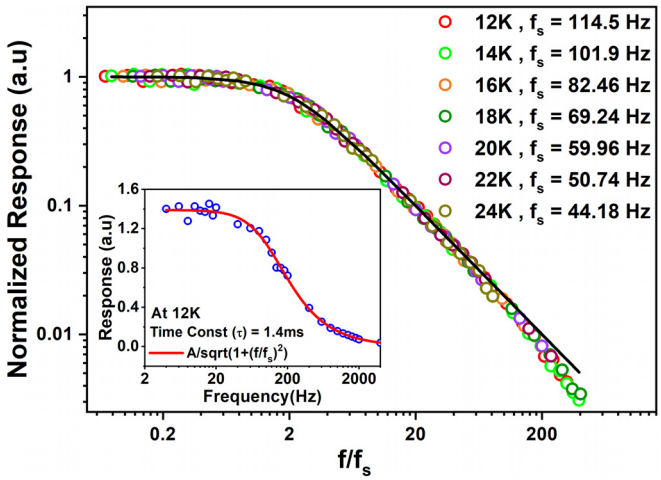

FIG. 3. Frequency scaled bolometer response at different bath temperatures fitted with a single pole roll off (black). Inset: Bolometer response as a function of heater frequency at 12 K.

heater signal is described by its heat capacity C and the thermal conductance G of the membrane legs[9] given by

$$G = \frac{\Delta H}{\Delta T}\bigg|_{T_b} = \frac{\partial H}{\partial \theta}\frac{\partial \theta}{\partial f_r}\frac{\partial f_r}{\partial T}. \tag{2}$$

To determine G from the above equation, the inverse responsivity $S_\theta^{-1} = \partial H/\partial\theta$ is calculated by measuring $\Delta\theta$ from the perturbation of the S_{21} signal away from the bias point [Fig. 2(d)].

The dependence of angular response $\Delta\theta$ on f_r is determined by plotting the resonant frequency at different bath temperatures, T_b, on the normalized unit circle in Fig. 2(d). The calculation of $\partial f_r/\partial T$ is described later. The thermal conductance is related to the heat capacity by the relation $\tau_s = C/G$, where τ_s is thermal time constant of the KIB for a given bath temperature. τ_s is determined from the cutoff

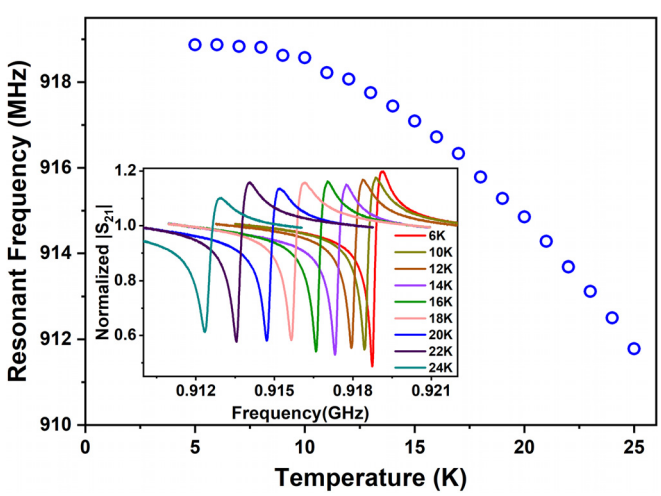

FIG. 4. Resonant frequency as a function of bath temperature, T_b. Inset: normalized S_{21} at different T_b values.

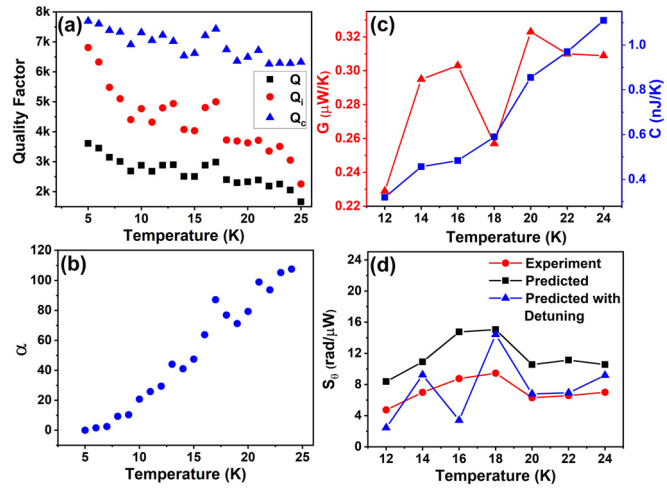

FIG. 5. Measured temperature dependence of resonator parameters: (a) Quality factor. (b) Temperature sensitivity, α. (c) Thermal conductance, G, and heat capacity, C. (d) Responsivity S_θ and predicted values based on measured model parameters including bolometer bias power, α, and detuning.

frequency of $f_s = 1/2\pi\tau_s$ of the AC response of the bolometer. The AC response of the bolometer is measured from the Fourier component of the mixed down transmission signal at twice the applied heater frequency. We measured f_s of the bolometer, biased at resonance for different bath temperatures, by applying a 3 V_{pp} voltage at frequencies ranging from 1 Hz to 1 kHz to the on-membrane heater. The frequency scaled heater responses for different bath temperatures all collapse to a single pole roll off function (Fig. 3). At $T_b = 12$ K, the measured thermal conductivity of the bolometer was $G = 0.229\ \mu$W/K at a resonant frequency of $f_r = 917.82$ MHz. The cut-off frequency at 12 K was $f_s = 114.5$ Hz, corresponding to a thermal response time of $\tau_s = 1.4$ ms. Having determined the thermal time constant, we calculate the heat capacity at 12 K as $C = \tau_s G = 0.32$ nJ/K.

Next, the bath temperature is slowly increased and S_{21} is measured. The shift in f_r as a function of T_b is plotted in Fig. 4. Both f_r and Q decrease with increasing temperature. By fitting the S_{21} data with Eq. (1), we determined f_r (Fig. 4) and Q [Fig. 5(a)] as functions of temperature. The temperature sensitivity of the KIBs, which depends on the resonance shift with the change in T_b and on Q_i, is commonly parametrized by the quantity α defined as[17]

$$\alpha = -2Q_i\frac{T}{f_r}\frac{\partial f_r}{\partial T}, \tag{3}$$

where $\partial f_r/\partial T$ is determined from Fig. 4 at different bath temperatures. α as a function of temperature is shown in Fig. 5(b).

We calculated C and G from 12 K to 24 K by applying a small AC signal to the on-membrane heater, as discussed above, which is shown in Fig. 5(c). At low temperatures, below 12 K, the KIB response to the heater signal was small and the resonance shift with temperature had an anomalous behavior as can be seen in Fig. 4. This resulted in inconsistent C and G measurements, and hence, data below 12 K are not shown.

The responsivity to low frequency thermal signals in the bolometer is estimated using the resonator-bolometer theory[17] as

Published under license by AIP Publishing

$$S_\theta = \frac{\partial\theta}{\partial H} \approx \frac{\alpha/GT}{1 + 2\alpha Q\frac{P_0}{GT}x + 4Q^2x^2}, \quad (4)$$

where $x = (f - f_r)/f_r$ is the detuning and P_0 is the power dissipated in the bolometer. (This approximation assumes a matched resonator and neglects the bolometer model parameters $Re[\alpha]$ and β, Ref. 17). We operated the detector such that about −56 dBm to −50 dBm is dissipated in the bolometer. At these low powers, the power dependent effects are small. Also, we chose a bias frequency, near the minimum of the transmission rather than precisely at the point of the highest S_θ, and so, $x \sim 1$. The measured and predicted responsivities, shown in Fig. 5(d), are in good agreement when the effect of de-tuning is taken into account.

The agreement between theory and experiment, together with our measurements of the bolometer's thermal properties, allows us to design better optimized YBCO bolometer arrays for the high temperature applications with the confidence that the technology is suitable for these goals. We estimate that a device on a 0.25 μm thick membrane would have a heat capacity of roughly $C = 1$ nJ/K at 55 K. As long as G does not rapidly increase with temperature as we go from 25 to 55 K, we can design membrane legs which are not too long, yet have a sufficiently low G. However, if our legs get longer, we can make room between pixels with innovative design of our spectrometer optics. We will design our membrane legs for the thermal conductance to be about $G = 0.04\,\mu$W/K with a thermal bandwidth of 60 Hz. This results in a NEP $= 3 \times 10^{-13}\,\mathrm{W}/\sqrt{\mathrm{Hz}}$ at 55 K. Currently, our noise measurements are limited by synthesizer phase noise of approximately −95 dBc. The theoretical phase noise of the bolometers is given by $N_{dBC} = 20\log(S_\theta P_{ph})$, which ranges from −125 dBc/Hz to an expected −140 dBc/Hz in 55 K devices, depending on the temperature, Q, α, G, and bias power. The synthesizer phase noise may be reduced to this level through the use of a stable reference clock for the synthesizer to achieve the optimal sensitivity of the bolometers.

This work was performed at the Jet Propulsion Laboratory (JPL), operated by the California Institute of Technology (Caltech) under Prime Contract No. NNN12AA01C with the National Aeronautics and Space Administration (NASA) through the Planetary Instrument Concepts for the Advancement of Solar System Observations (PICASSO) Program. MgO deposition was performed by I-Beam Materials, Santa Fe, NM. YBCO and Au were deposited by Ceraco Ceramic Coating, Ismaning, Germany. Etching of those layers was performed by STAR Cryoelectronics, Santa Fe, NM.

REFERENCES

[1] P. Day, H. LeDuc, B. Mazin, A. Vayonakis, and J. Zmuidzinas, Nature **425**, 817 (2003).
[2] B. A. Mazin, B. Bumble, S. R. Meeker, K. O'Brien, S. McHugh, and E. Langman, Opt. Express **20**, 1503 (2012).
[3] S. McHugh, B. A. Mazin, B. Serfass, S. Meeker, K. O'Brien, R. Duan, R. Raffanti, and D. Werthimer, Rev. Sci. Instrum. **83**, 044702 (2012).
[4] J. Van Rantwijk, M. Grim, D. Van Loon, S. Yates, A. Baryshev, and J. Baselmans, IEEE Trans. Microwave Theory Tech. **64**, 1876 (2016).
[5] G. Ulbricht, B. A. Mazin, P. Szypryt, A. B. Walter, C. Bockstiegel, and B. Bumble, Appl. Phys. Lett. **106**, 251103 (2015).
[6] P. Richards, J. Clarke, R. Leoni, P. Lerch, S. Verghese, M. R. Beasley, T. H. Geballe, R. H. Hammond, P. Rosenthal, and S. R. Spielman, Appl. Phys. Lett. **54**, 283 (1989).
[7] S. J. Berkowitz, A. S. Hirahara, K. Char, and E. N. Grossman, Appl. Phys. Lett. **69**, 2125 (1996).
[8] B. A. Steinbach, J. J. Bock, H. T. Nguyen, R. C. O'Brient, and A. D. Turner, J. Low Temp. Phys. **193**, 88 (2018).
[9] J. Mather, Appl. Opt. **21**, 1125 (1982).
[10] M. S. Khalil, M. J. A. Stoutimore, F. C. Wellstood, and K. D. Osborn, J. Appl. Phys. **111**, 054510 (2012).
[11] S. Probst, F. B. Song, P. A. Bushev, A. V. Ustinov, and M. Weides, Rev. Sci. Instrum. **86**, 024706 (2015).
[12] M. A. Lindeman, J. A. Bonetti, B. Bumble, P. K. Day, B. H. Eom, W. A. Holmes, and A. W. Kleinsasser, J. Appl. Phys. **115**, 234509 (2014).
[13] F. Flasar and V. Kunde, Space Sci. Rev. **115**, 169 (2004).
[14] R. Brown, K. Baines, and G. Bellucci, Space Sci. Rev. **115**, 111 (2004).
[15] V. Cottini, C. A. Nixon, D. E. Jennings, R. de Kok, N. A. Teanby, P. G. J. Irwin, and F. M. Flasar, Planet. Space Sci. **60**, 62 (2012).
[16] J. M. Phillips, J. Appl. Phys. **79**, 1829 (1996).
[17] M. A. Lindeman, J. Appl. Phys. **116**, 024506 (2014).

Published under license by AIP Publishing